AI Assignment -- Classical AI Reasoning System

Formal First-Order Logic Representation, Robinson's Unification,
Chaining Algorithms, and Structural Causal Counterfactual Inference

STUDENT SUBMISSION DETAILS

Roll Number: M26CSA013

Artificial Intelligence (CSL7610)

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Executive Project Summary

This submission presents a fully functional, general-purpose classical automated reasoning system designed to decide logical entailment over natural language theories, formal logic formulas, and counterfactual causal scenarios. Strictly conforming to course guidelines, the inference pipeline does not utilize any generative LLM or black-box neural classifier. All deductive proofs are conducted deterministically through custom-built algorithms: Agenda-based Forward Chaining, Goal-directed SLD Backward Chaining, Robinson's First-Order Unification, and Pearl's Structural Causal Counterfactual Abduction-Intervention-Deduction Engine.

Comprehensive empirical evaluations were conducted on three benchmark datasets from Hugging Face:

1. RuleTaker (tasksource/ruletaker) -- Multi-step Horn-clause deduction
2. FOLIO (tasksource/folio) -- First-order logic theorem proving with formal annotations
3. CounterBench (CounterBench/CounterBench) -- Deterministic counterfactual causal reasoning

Additionally, an automated suite of 23 unit tests was executed with a 100% pass rate.

1. Assignment Objective and Architectural Pipeline

The objective is to implement an end-to-end reasoning agent that takes a natural-language knowledge base (facts and rules) and a query, and outputs a binary True/False entailment prediction. The system strictly follows the modular pipeline: Natural Language / Formal Annotations -> Internal Logical Representation -> Classical Automated Reasoning Engine -> Entailment Prediction. No machine-learning classifier or generative language model is used for the final entailment decision.

2. Problem Formulation & Logical Semantics

Let Delta be a formal knowledge base containing ground atomic facts F and rules R. Let Q be a ground literal query. The inference engine determines whether Delta |= Q. We adopt the Closed World Assumption (CWA) and Negation-as-Failure (NAF): if Q cannot be derived through sound inference rules, it evaluates to False. Non-entailment is explicitly distinguished from proof of negation. For counterfactual queries, we implement Judea Pearl's Structural Causal Model (SCM) logic: abduction of factual observations, graph surgery / intervention do(X = x), and deterministic forward constraint propagation.

3. Dataset Inspection, Loading, and Preparation

Three official public Hugging Face benchmark datasets were loaded and evaluated:
- RuleTaker (tasksource/ruletaker): Contains synthetic English theories and facts with depth-bounded reasoning (depth-0 to depth-5). Evaluated on the official dev split with labels 'entailment' (True) and 'not entailment' (False).
- FOLIO (tasksource/folio / yale-nlp/folio): Expert-annotated first-order logic reasoning benchmark with premises, conclusions, and formal Unicode FOL formulas (premises-FOL, conclusion-FOL). Labels: True, False, and Uncertain. Uncertain instances are excluded from binary accuracy computation and predicted as False under CWA.
- CounterBench (CounterBench/CounterBench): Comprises 1.2K counterfactual reasoning questions across two files: 'data_balanced_alpha_V1.json' (1,000 items) and 'data_balanced_backdoor_V2.json' (200 items). Individual JSON files were loaded and validated, mapping 'given_info' to context, 'question' to query, and 'answer' ('yes'/'no') to boolean labels. Model definitions ('meta_model_alpha_V1.json', 'meta_model_backdoor_V2.json') were kept distinct.

4. Internal Logical Representation

The system defines strongly-typed, immutable data structures in Python:
- Literal: An atom consisting of a predicate functor, ordered term arguments (constants or logical variables), and a negation flag. Variables begin with uppercase letters (e.g., X, Y, V) or '?', while constants are lowercase identifiers.
- Clause: A Horn clause representing Head :- Body_1 and ... and Body_k. An empty body represents a ground fact.
- KnowledgeBase: Stores facts and rules, indexed by (predicate, arity) for O(1) candidate matching.

5. Natural-Language, FOL, and Counterfactual Parsers

Parsing is partitioned into three specialized modular components:
1. Deterministic NL Parser (src/parser.py): Converts natural-language English sentences into facts and rules using regex patterns. Handles is-a classifications ('Bob is a dog'), attribution ('Dave is young and kind'), negations ('Anne is not round'), universal rules ('All rough things are quiet'), multi-premise rules ('Kind, young things are smart'), transitive relations ('The cat chases the mouse'), and anaphoric pronoun bindings ('they are...').
2. Formal FOL Parser (src/folio_fol_parser.py): Converts Unicode first-order logic annotations from FOLIO into internal clauses.
3. Counterfactual SCM Parser (src/counterfactual_engine.py): Extracts causal relations ('X causes Y', 'A and B together cause C', 'A or B causes C'), interventions ('if not X instead of X', 'suppose not Y'), and observations ('We observed T').

6. Robinson's First-Order Unification

The unification engine (src/unification.py) computes the Most General Unifier (MGU):
- Variable-to-constant binding: binds variable X to constant c, updating substitution {X: c}.
- Variable-to-variable binding: maintains equality substitution classes.
- Repeated variable constraints: verifies consistency across duplicate variables (e.g. same(X, X)).
- Occurs-check: prevents circular structures.
- Standardisation apart: appends unique integer suffixes (_1, _2, ...) to rule variables during derivation to avoid variable capture.

7. Classical Reasoning Algorithms

Three distinct reasoning engines are implemented:

A. Goal-Directed Backward Chaining (SLD Resolution): Recursively proves query subgoals with cycle detection and depth bounding (max_depth=50).

B. Agenda-Based Forward Chaining: Iteratively derives new ground facts until a fixpoint or query derivation.

C. Counterfactual Structural Causal Engine: Executes Pearl's 3-step SCM deduction: (1) Abduction of observed factual states, (2) Graph surgery to fix intervened variables and sever incoming causes, and (3) Forward constraint propagation across causal equations.

8. Detailed Step-by-Step Reasoning Traces

A. Assignment Specification Benchmark Example

```
Facts:  Young(bob), Kind(bob)
Rules:  Smart(X) :- Young(X)
        Successful(X) :- Smart(X), Kind(X)
Query:  Successful(bob)

Execution Trace (Backward Chaining):
  1. Goal: successful(bob)
  2. Match Rule: Successful(X) :- Smart(X), Kind(X) with subst {X_1: bob}
  3. Subgoal 1: smart(bob)
     - Match Rule: Smart(X) :- Young(X) with subst {X_2: bob}
     - Subgoal 1.1: young(bob) matches known fact -> PROVED
  4. Subgoal 2: kind(bob) matches known fact -> PROVED
  5. Conclusion: successful(bob) is logically entailed -> Prediction: True
```

B. Counterfactual Causal Deduction Example (CounterBench)

```
Given:  Ziklo causes not Blaf, Blaf causes Lumbo.
Query:  Would Lumbo occur if not Ziklo instead of Ziklo?

Execution Trace (Pearl SCM Counterfactual Engine):
  1. Target variable: lumbo
  2. Graph Surgery (Intervention): do(ziklo = False)
  3. Rule evaluation: ziklo -> not blaf. Since ziklo=False, blaf = True.
  4. Rule evaluation: blaf -> lumbo. Since blaf=True, lumbo = True.
  5. Conclusion: lumbo evaluates to True in counterfactual world -> Prediction: True (yes)
```

9. Empirical Evaluation and Measured Results

All experiments were executed on public datasets from Hugging Face using high-resolution timers (time.perf_counter()). Accuracy, Precision, Recall, F1, and Confusion Matrices were computed rigorously in pure Python without synthetic data:

Benchmark	Split	N (eval)	Accuracy	Precision	Recall	F1	Parse Fail%
RuleTaker	dev	200	58.50%	0.6019	0.5962	0.5990	0.0%
CounterBench	alpha	1000	89.00%	0.8920	0.8880	0.8900	0.0%
CounterBench	backdoor	200	62.50%	0.5820	0.7470	0.6540	0.0%
FOLIO	validation	134*	50.75%	0.6000	0.2500	0.3529	19.2%
Unit Tests	suite	23	100.0%	1.0000	1.0000	1.0000	0.0%

*Note: In FOLIO, 69 uncertain examples were excluded from binary accuracy computation per assignment instructions.

Inference Timing Profiling (High-Resolution per Query)

Benchmark	Parsing Time	Reasoning Time	Total Time / Query
RuleTaker	0.140 ms	0.035 ms	0.345 ms
CounterBench (Alpha)	0.030 ms	0.048 ms	0.084 ms
CounterBench (Backdoor)	0.025 ms	0.028 ms	0.055 ms
FOLIO	0.140 ms	0.017 ms	0.249 ms

10. Comparative Analysis: Backward vs. Forward vs. Causal Chaining

- Backward Chaining: Ideal for goal-directed queries (0.009-0.015 ms), searching only relevant proof trees.
- Forward Chaining: Well-suited for exhaustive fact derivation over static rule bases, but incurs higher premise matching cost.
- Counterfactual SCM Engine: Essential for intervened causal reasoning where graph surgery modifies rule firing dynamically.

11. Error Analysis and Theoretical Limitations

1. Stochastic Backdoor Adjustments: CounterBench backdoor examples contain Bernoulli distributions (e.g. Bern(0.2)).

Deterministic boolean propagation handles causal paths but cannot perform probabilistic Bayesian conditioning.
2. Disjunctive Rule Heads: Full First-Order Logic allows disjunctive consequents (A -> B v C), which require full first-order resolution theorem proving rather than Horn-clause chaining.
3. Open-World vs. CWA Semantics: In FOLIO, 'Uncertain' hypotheses are classified as False under CWA, introducing false negatives.

12. Deliverable Verification Checklist

All four required deliverables are verified and placed in the submission/ directory:
- reasoning_agent.ipynb: Self-contained, fully executed 20-section notebook.
- report.pdf: This comprehensive technical report with Student Roll Number M26CSA013.
- submission.csv: Validated predictions table containing 'id' and 'predicted_label'.
- README.txt: Complete environment, execution, and reproducibility instructions.

13. References

[1] Clark, P., Tafjord, O., & Richardson, K. (2020). Transformers as Soft Reasoners over Language. IJCAI-PRICAI 2020.
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[5] Robinson, J. A. (1965). A Machine-Oriented Logic Based on the Resolution Principle. JACM, 12(1), 23-41.
[6] Russell, S., & Norvig, P. (2020). Artificial Intelligence: A Modern Approach (4th ed.). Pearson.